PROJECT 3 | RECOMMENDATION MEMO

Results from our periodic NLS test

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We followed your suggestion and tested the periodic cubic nonlinear Schrodinger equation against a Fourier reference, using three random seeds.

The default PINN's median error rose from about 0.09% at T=1 to 110% at T=4. The same network fitted to reference data reached 0.76% at T=4. Starting physics-only training from that fit gave 43%. The same architecture fits the field much more accurately with supervision, so the way we train with physics matters in this test.

Enforcing mass accurately did not keep the predicted wave accurate: large phase and profile errors remained. Larger penalties and more time samples also failed to give a reliable longer-time solution.

For this single forward problem, we would keep the Fourier solver. It was fast and much more accurate. We have not shown a practical benefit from replacing it with the tested PINNs.

Next, we would test shorter time windows, passing each window's final state to the next, and compare solution error, phase, and Hamiltonian drift at matched cost. We have not tested that remedy yet.

We are confident in this comparison for our test case. The conclusion is limited to this initial condition, architecture, three seeds, and training budget.